Northeastern University
Bouvé College *of* Health Sciences

Course number: PHTH 6830

Course name: Generalized Linear Models

Fall 2025

Instructor

Instructor: Larry Han, Assistant Professor, Department of Public Health and Health Sciences

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Office: 316D Robinson Hall

Office Hours: Thursdays 12:00 – 1:00 pm, Robinson Hall, Room 316D

Course Information

Class meeting time: Thursdays 1:35 – 4:55 pm, Kariotis Hall, Room 204

Description: Focuses on selecting, fitting, and evaluating the general class of generalized linear models. Emphasizes linear, logistic, Poisson, and proportional hazards regression in addition to time series analyses. Stresses proper construction of models through the evaluation of modeling assumptions and assessment of model diagnostics. Introduces concepts of confounding, effect modification, and methods for missing data. Offers students an opportunity to fit, evaluate, and appropriately communicate and visualize the results of modeling techniques using the R programming language and statistical software packages, as well as practice on techniques for data cleaning on real examples of messy data.

Prerequisites: None

Corequisites: None

Credit Hours: 4

Objectives / Outcomes

After completion of this course, students will be able to:

- Understand and describe linear regression for analyzing continuous response data, and apply it to analyze data.
- Understand and describe the class of generalized linear models, and apply them to analyze data.
- Analyze continuous and discrete response data, including data on binary, count, categorical, and survival outcomes.

Teaching Methods / Learning Experiences

This course will feature a combination of (i) lectures presented by the instructor to introduce new course material, through slides, whiteboard, videos, and course notes, (ii) readings, presentations, and discussion of classical and modern papers, both by the instructor and by the

students themselves, and (iii) problem sets and labs designed by the instructor for students to gain hands-on experience solving problems and using software (primarily through R). Throughout the semester, students will receive feedback from the instructor on their progress. Students will have the opportunity to work together on problem sets.

Grading / Evaluation

Grades will be determined by the following four components:

- Homework: 50%
  - Homework will be assigned approximately every one to two weeks. Students may work on homework together and learn from each other, but the write up of solutions should reflect each student's own understanding.
- Mid-term exam: 25%
- Final exam: 25%
  - Both the midterm and final exams will be in-class. You will be allowed a single-sided, standard sheet of notes to help you.

Note: Grades will be rounded to the nearest tenth. For instance, an 79.6% → 80%

*See University Grading Scales for UG and Grad here:
https://registrar.northeastern.edu/article/university-grading-system/

Materials / Resources

Electronic copies of lecture notes, slides, supplementary materials, homework assignments, datasets, code, and labs will be posted on the course Canvas. While there are no required texts, the following are good references.

- McCullagh P, Nelder J. Generalized Linear Models (1989)
- Collet D. Modeling Survival Data in Medical Research, 2nd Edition. Chapman-Hall, 2003)
- Faraway J. Extending the Linear Model with R (2006)
- Hastie T, Tibshirani R, Friedman J. Elements of Statistical Learning, 2nd Edition (2009)
- Agresti A. Categorical Data Analysis, 3rd Edition (2012)
- Gelman A, Carlin, J, Stern H, Dunson D, Aki V, Rubin D. Bayesian Data Analysis, 2rd Edition (2013)
- Wakefield J. Bayesian and Frequentist Regression Methods (2013).

Canvas: https://canvas.northeastern.edu

Course Policies

Student Expectations:

If you are unable to attend class, please let me know in advance, if possible. For prolonged absences, contact me to discuss a make-up plan.

Class Participation:

This is a participatory course and each student is expected to be an active participant in the work that occurs. Student learning in this course is directly related to the efforts made in learning the course material. Attending class and arriving to class promptly is expected of each student (and shows respect for fellow students). Students will be called on in class, and each student can expect to be called on unexpectedly several times during the semester.

Professional Conduct:

Part of the student learning experience is to practice professional behavior. This includes affording your student peers, presenters and professors the courtesy of not using computers, cell phone, pagers or text messaging during class. Non-adherence to these expectations of professional behavior will result in dismissal from class. Consistent with University policy, students must seek the course instructor's permission to record class lectures.

Late Work Policy:

You are allowed to turn in one problem set a maximum of 48 hours past the due date. Please notify the instructor by email if you intend to use your one "late pass". Otherwise, no credit will be given for late assignments unless prior arrangements have been made with the instructor.

Academic Integrity:

As health professionals and scientists-in-training, the highest level of ethical behavior is expected of all students. Plagiarism, cheating, and any form of unauthorized collaboration will not be tolerated and will be handled in accordance with University policies including the Academic Honesty and Integrity Policy. Requirements can be found at:
https://osccr.sites.northeastern.edu/academic-integrity-policy/

https://osccr.sites.northeastern.edu/code-of-student-conduct/

- Acts of academic dishonesty will be referred to OSCCR (Office of Student Conduct and Conflict Resolution).
- Infractions will be dealt with according to Northeastern Disciplinary Process shown in http://www.northeastern.edu/osccr/academic-integrity-report-form/
- The Turnitin application may be used by the course instructor to identify and document cases of plagiarism.

Use of Generative Artificial Intelligence (AI)

Generative AI tools should not be used to write full solutions to problem sets. You are expected to work through each problem on your own first, as this process is essential for building understanding and critical thinking. I encourage you to use AI tools to check your work—particularly for coding—but not to rely on them, since doing so will shortchange your learning and prevent you from developing the skills needed by the end of the course.

Belonging at Bouve' College:

We acknowledge that chronic and persistent disparities in education and health due to systemic oppression, based on race, class, and other marginalized identities, have adversely affected individuals and society. At Bouve´ we are committed to educating future healthcare leaders who strive to serve marginalized communities and improve the well-being of all members of society through the health span.

Within the college, we strive to create a safe environment for diverse thoughts, perspectives, and experiences to be shared, explored, and embraced. As a community we recognize that disrupting long-standing oppressive practices is a process. Everyone is at a different place on this journey, and we affirm our commitment to support and learn from one another.

In order for everyone in our learning community to grow, gain a deeper understanding of complex issues and be exposed to diversity of thought, we may have discussions in this class on topics that could be uncomfortable or difficult for some people. It is imperative that we follow rules of engagement as below in order to create a psychologically safe learning environment.

Ground Rules:

1. Listen actively and respect others when they are talking.
2. Participate to the fullest of your ability without taking up too much space – community growth depends on the inclusion of every individual voice
3. Speak from your own experience instead of generalizing ("I" instead of "we")
4. Do not be afraid to respectfully challenge one another by asking questions but refrain from personal attacks and stay focused on ideas rather than individuals
5. Be conscious of body language and nonverbal responses; they can be as disrespectful and hurtful as words.

Finally, all community members need to adhere to the NU Student Code of Conduct "e*ach Northeastern student has the responsibility to respect the differences of individuals and treat others in a civil and respectful manner."*

All class and group work interactions are expected to be courteous and respectful of other's health and cultural beliefs and opinions. I consider it a part of my responsibilities as instructor to address the learning needs of all students in this course. I also believe that the diversity of student experiences and perspectives is essential to the deepening of knowledge in this course. Any suggestions that you have about other ways to include the value of diversity in this course are welcome.

Belonging at Northeastern

Students with Disabilities:

Disability Access Services (DAS), located on campus in 10 Dodge Hall (617-373-2675) can provide students with information and other assistance to help manage any challenges that may affect their performance in the coursework. The University requires that students provide documentation of their disability to the DAS. Students should meet with the course instructor for special accommodations to be arranged.

Northeastern University abides by Section 504 of the Rehabilitation Act of 1973, which stipulates that no student shall be denied the benefits of an education "solely by reason of a

handicap." Disabilities covered by law include but are not limited to, learning disabilities and hearing, sight or mobility impairments. Additional information about DAS can be found online at https://drc.sites.northeastern.edu/

Student Wellness:

As health researchers, we do our best to maintain a healthy, balanced life. As a student however, you may experience a range of challenges including significant stress, difficult life events, mood changes, excessive worry, or problems with eating and/or sleeping. These can diminish your academic performance and/or reduce your ability to participate in daily activities. If you or anyone you know is struggling, we strongly encourage you to seek support. Northeastern University provides several services and resources to support the overall wellness of students:

University Health and Counseling Services

Find at Northeastern

We Care

Wellness Days

As always, please contact Northeastern University Police Department at 617-373-3333 in emergency situations.

Title IX Policy:

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including students, faculty and staff of all gender identities.

Learn more at:

https://www.northeastern.edu/ouec/training-and-education/syllabus-language-title-ix/

Know Your Options: Pregnancy

Course Feedback:

The Office of the Registrar, in collaboration with the Office of the Provost, administers the TRACE (Teacher Rating and Course Evaluation) for undergraduate and graduate courses at Northeastern University. Students' assessments through TRACE are completely anonymous. Given the public nature of student comments made through TRACE and their intended use to enhance teaching, students are asked to please be thoughtful, professional and considerate with responses.

Syllabus Disclaimer:

Any changes to the syllabus will be made at the discretion of the instructor. It is the responsibility of the student to stay informed of any such changes.

For additional information, resources, and policies please visit the Northeastern University Catalog* including:

- University Undergraduate and University-wide academic policies and procedures
- University Graduate and University-wide academic policies and procedures
- Bouvé College of Health Sciences Undergraduate and Graduate policies, and program-specific policies and procedures

**Students should consult the appropriate version based on enrollment of the Northeastern academic catalogs, their academic advisors, and/or their degree audit reports for their specific requirements.*

Major Assignment / Course Schedule

Weeks/Dates	Topics/Assignments/Assessments/etc.	Due Dates
1 – Sep 4	Unit 1 – Heteroskedastic continuous response data	
2 – Sep 11	Unit 1 – Heteroskedastic continuous response data HW 1 Assigned	
3 – Sep 18	Unit 2 – Generalized linear models	
4 – Sep 25	Unit 2 – Generalized linear models HW 2 Assigned	HW 1 due
5 – Oct 2	Unit 3 – Analysis of binary response data	
6 – Oct 9	Unit 3 – Analysis of binary response data HW 3 Assigned	HW 2 due
7 – Oct 16	Unit 4 – Analysis of count response data	
8 – Oct 23	Unit 4 – Analysis of count response data HW 4 Assigned Midterm Exam	HW 3 due
9 – Oct 30	Unit 5 – Analysis of multinomial response data	
10 – Nov 6	Unit 5 – Analysis of multinomial response data	HW 4 due

Weeks/Dates	Topics/Assignments/Assessments/etc.	Due Dates
	HW 5 Assigned	
11 – Nov 13	Unit 6 – Analysis of time-to-event response data	
12 – Nov 20	Unit 6 – Analysis of time-to-event response data HW 6 Assigned	HW 5 due
13 – Nov 27	Thanksgiving – No Class	
14 – Dec 4	Unit 7 – An introduction to causal inference	HW 6 due
15 – Dec 11	Final Exam	